Functions: Composite & Inverse

Section 1: Domain, Range & Function Concepts

1 Find the natural domain and range of the function $f(x) = \sqrt{2x - 6}$.

a. Domain: $x \geq 3$; Range: $f(x) \geq 0$

b. Domain: $x > 3$; Range: $f(x) > 0$

c. Domain: $x \geq 0$; Range: $f(x) \geq -6$

d. Domain: $x \in \mathbb{R}$; Range: $f(x) \geq 0$

2 The function $g(x) = x^2 - 4x + 7$ is defined for $x \in \mathbb{R}$. What is the range of g?

a. $g(x) \geq 3$

b. $g(x) \geq 7$

c. $g(x) \geq -3$

d. $g(x) \in \mathbb{R}$

3 Explain why the function $h(x) = x^2 - 4$ for $x \in \mathbb{R}$ does not have an inverse.

a. It is a many-to-one function on $\mathbb{R}$

b. Its domain does not include negative numbers

c. Its range is bounded below by -4

d. It is not continuous at $x = 0$

Section 2: Composite Functions

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4 Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find the expression for $fg(x)$.

a. $2x^2 + 1$

b. $4x^2 + 12x + 8$

c. $2x^2 + 5$

d. $2x^2 - 2$

5 Given $f(x) = \dfrac{1}{x+2}$ for $x \neq -2$ and $g(x) = 3x - 1$, solve the equation $gf(x) = 1$.

a. $x = 1/2$

b. $x = -1$

c. $x = 2$

d. $x = 0$

6 If $f(x) = 3x - 4$, find the expression for $ff(x)$.

a. $9x - 16$

b. $9x - 12$

c. $6x - 8$

d. $9x^2 - 24x + 16$

Section 3: Inverse Functions & Self-Inverses

7 Find the inverse function $f^{-1}(x)$ for $f(x) = \dfrac{2x-5}{x+1}$ $(x \neq -1)$.

a. $f^{-1}(x) = \dfrac{x+5}{2-x}$

b. $f^{-1}(x) = \dfrac{x-5}{2-x}$

c. $f^{-1}(x) = \dfrac{2x+1}{x-5}$

d. $f^{-1}(x) = \dfrac{x+1}{2x-5}$

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8 The function $f(x) = x^2 - 6x + 10$ is defined for $x \geq 3$. Find the expression for $f^{-1}(x)$.

a. $f^{-1}(x) = 3 + \sqrt{x - 1}$ for $x \geq 1$

b. $f^{-1}(x) = 3 - \sqrt{x - 1}$ for $x \geq 1$

c. $f^{-1}(x) = \sqrt{x + 6} - 3$

d. $f^{-1}(x) = 3 + \sqrt{x + 1}$

9 A function $g(x) = \dfrac{ax + b}{cx + d}$ is a self-inverse function ($g(x) = g^{-1}(x)$). Which condition guarantees this property for all valid x?

a. $a + d = 0$

b. $ad - bc = 1$

c. $a = d$

d. $b = c$

Section 4: Graphical Relationships & Exam Challenges

10 What is the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$?

a. Reflection in the line $y = x$

b. Reflection in the x-axis

c. Reflection in the y-axis

d. Rotation of 180° about the origin

11 Solve the equation $f(x) = f^{-1}(x)$ where $f(x) = 2x - 3$.

a. $x = 3$

b. $x = 1$

c. $x = -3$

d. $x = 2$

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12 The function $f(x) = 2x^2 - 8x + c$ has domain $x \geq k$. Find the least value of k for which f has an inverse.

a. $k = 2$

b. $k = 4$

c. $k = 0$

d. $k = -2$

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